AMATH 582: HW 1

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Abstract. This report investigates a methodology for tracking the trajectory of a submarine utilizing data derived from a sparse array of underwater sensors. The primary objective is to isolate the target's dominant center frequency and design a spectral filter to extrapolate the submarine's position over time. To achieve this, the raw sensor data is transformed into Fourier space to facilitate efficient processing and spectral analysis. Following the design and application of a frequency-domain filter, we reconstruct the movement of the submarine across a 24-hour dataset. The results demonstrate that we were able to identify the target's trajectory to enable continued aerial monitoring.

1. Introduction and Overview

We consider a discretized spatial domain within the Puget Sound, modeled as a uniformly distributed grid. The domain is defined by 64 grid points in each spatial dimension (x, y, z), equipped with hydroacoustic sensors. The system records a temporal sample at 30-minute intervals. While these sensors are tuned to a specific frequency bandwidth, they are susceptible to environmental noise and interference. This poses a significant signal processing challenge, as the primary objective is the early detection and continuous tracking of unauthorized vessels. It is imperative to identify and isolate the specific spectral frequencies corresponding to an intruder. However, to achieve accurate detection, the signal of interest must be decoupled from the 'background' acoustic floor, which includes noise pollution from commercial ships, military sonar, and biological activity [1].

1.1. Mathematical Formulation.

The data collected by the hydrophone array can be represented as a four-dimensional tensor $\mathbf{D} \in \mathbb{R}^{n_x \times n_y \times n_z \times T}$, where $n_x = n_y = n_z = 64$ represents the spatial resolution and T represents the temporal dimension. For a given spatial coordinate $\mathbf{x} = (x, y, z)$ and time t, the recorded acoustic pressure $P(\mathbf{x}, t)$ is modeled as a sum of the target source $S(\mathbf{x}, t)$ and background noise $\eta(\mathbf{x}, t)$:

$$P(\mathbf{x}, t) = S(\mathbf{x}, t) + \eta(\mathbf{x}, t) \tag{1}$$

The noise term $\eta(\mathbf{x}, t)$ is assumed to be a combination of Gaussian white noise and structured interference from known marine sources. The objective is to determine a transfer function or filter $\mathcal{F}$ such that $||\mathcal{F}(P) - S||$ is minimized, thereby isolating the submarine's frequency and therefore its trajectory.

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2. Theoretical Background

To isolate the submarine's trajectory from the noisy hydrophone data, we rely on spectral analysis techniques. This section outlines the nature of the noise, the mathematical foundations of the Fourier transformation, and the computational algorithms required to process high-dimensional data efficiently.

2.1. **Assumed Noise.** We model the acoustic environment as a sum of a deterministic signal $S(\mathbf{x}, t)$ and a noise term $\eta(\mathbf{x}, t)$. The noise is assumed to be additive Gaussian white noise. In the spatial domain, this noise is distributed across all points, often obscuring the target signal. However, it is known that 'adding mean zero white noise to a signal is equivalent to adding mean zero white noise to its Fourier series coefficients' [2]. In contrast, the mechanical signature of a submarine is typically static or nearly so, allowing us to attempt to separate the two signals.

2.2. **Fourier Series.** The foundation of our analysis is the Fourier Series, which states that any periodic function $f(x)$ on a domain $[-L, L]$ can be represented as an infinite sum of complex exponentials. For a function $f(x)$, the series is defined as:

$$f(x) = \sum_{k=-\infty}^{\infty} c_k e^{i\frac{k\pi x}{L}} \tag{2}$$

where the coefficients c_k are determined by projecting the function onto the orthogonal basis functions:

$$c_k = \frac{1}{2L} \int_{-L}^{L} f(x) e^{-i\frac{k\pi x}{L}}\, dx \tag{3}$$

These coefficients c_k encode the amplitude and phase of the signal at the discrete frequency indexed by k. This concept implies that complex acoustic signals can be decomposed into discrete frequencies, allowing us to identify the specific "note" or center frequency emitted by the submarine's propulsion system.

2.3. **Fourier Transform.** Extending the Fourier Series to non-periodic functions over an infinite domain yields the continuous Fourier Transform. For our spatial signal $u(\mathbf{x})$, the 3-dimensional Fourier transform $\hat{u}(\mathbf{k})$ is defined as:

$$\hat{u}(\mathbf{k}) = \int_{\mathbb{R}^3} u(\mathbf{x}) e^{-i\mathbf{k}\cdot\mathbf{x}}\, d\mathbf{x} \tag{4}$$

where $\mathbf{k} = (k_x, k_y, k_z)$ represents the wavenumber vector. This transformation maps the physical location of sound (spatial domain) to the frequency of the sound (spectral domain). A submarine in 3D space may be difficult to find, but if it emits a nearly constant tone, it will appear as a distinct spike in the frequency domain. Since the Fourier Transform is a linear operator, the white noise we explored in Section 2.1 is additive in the frequency space as well. This allows us to filter the noise out while in the frequency domain and then transform it back to the spatial domain resulting in a cleaner signal.

2.4. **Frequency Domain Filtering.** Once the data is transformed into the frequency domain, filtering becomes a straightforward algebraic operation due to the independence of the noise and the linearity of the Fourier Transform. To isolate the submarine, we apply a filter $G(\mathbf{k})$, typically a Gaussian, centered around the target's dominant frequency $\mathbf{k}_{sub}$:

$$\hat{u}_{filtered}(\mathbf{k}) = \hat{u}(\mathbf{k}) \cdot G(\mathbf{k}) \tag{5}$$

By zeroing out frequencies where $|G(\mathbf{k})| \approx 0$, we effectively remove the white noise while preserving the frequencies of the submarine. An inverse Fourier transform is then applied to recover the denoised spatial coordinates.

2.5. **Computational Value of the FFT.** Calculating the Discrete Fourier Transform (DFT) directly for a grid of size N requires $\mathcal{O}(N^2)$ operations. Given our domain of $64 \times 64 \times 64$ points per time step, a direct calculation would be computationally expensive. To solve this, we utilize the Fast Fourier Transform (FFT), specifically the Cooley-Tukey algorithm. The FFT exploits the symmetries of the complex roots of unity to reduce the complexity from $\mathcal{O}(N^2)$ to $\mathcal{O}(N \log N)$ [3] [4].

$$\text{Speedup Factor} \approx \frac{N^2}{N \log_2 N} = \frac{N}{\log_2 N} \tag{6}$$

For $N = 64^3 \approx 2.6 \times 10^5$, the FFT is several orders of magnitude faster than the naive DFT, making real-time tracking of the submarine feasible.

3. Algorithm Implementation and Development

The high level process follows the following flowchart:

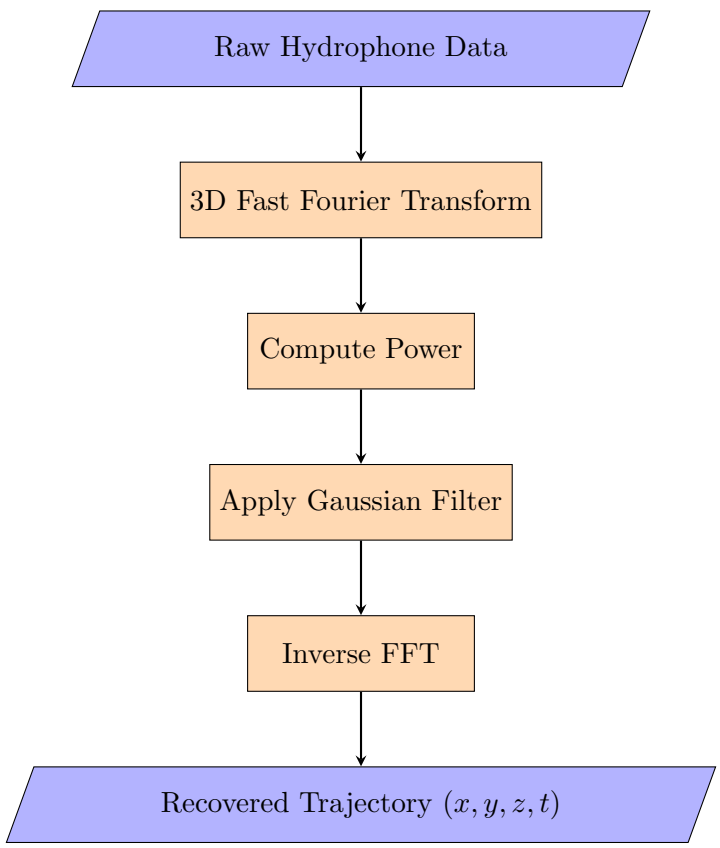

Figure 1. Signal processing pipeline for submarine localization.

(1) **Plot Noisy Data**
We initially plot a Z-max projection of the data. This can be thought of as a top down view of the water where we can see to the bottom of the area without issue. This gives us a starting point and also develops the idea of a projection of max data onto a particular dimension, an idea we will continue to rely on for visualizations.

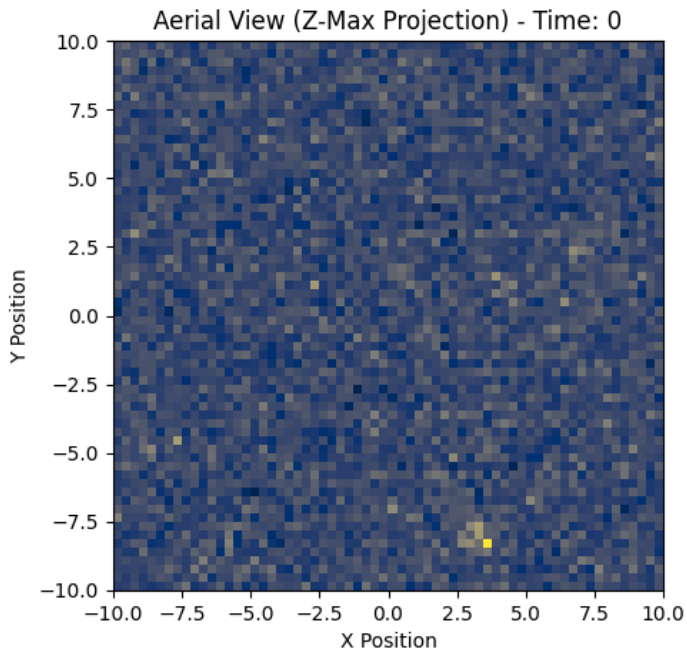

FIGURE 2. It is possible to identify the path of the submarine in the z-max projection but it is obfuscated by noise. The goal is to eliminate noise resulting in a clear picture of the submarine's path.

(2) **Compute 3D FFT and Plot**
To find the strength of a signal, we need to find the magnitude of the result from the FFT which is in the format $a + bi$. Formally, this is equal to $\sqrt{a^2 + b^2}$ but we can accomplish the same thing with the NumPy function `np.abs()`. Now we have the combined strength of the signals, we can further identify the power of the signal by squaring each dimension according to Parseval's Theorem [2].

$$\sum_{n=0}^{N-1} |x[n]|^2 = \frac{1}{N} \sum_{k=0}^{N-1} |X[k]|^2 \tag{7}$$

(3) **Calculate Projections onto each Spatial Dimensional Plane**
When we project these results on their corresponding $k_{x,y,z}$ planes, there is not much to differentiate the sub from the noise, as seen in Figure 3. From here we can refer back to the project statement and a note we introduced in 2.1 which we rely on to further differentiate strong signals from weak noise by dividing by the average of the power across the frequency domain.

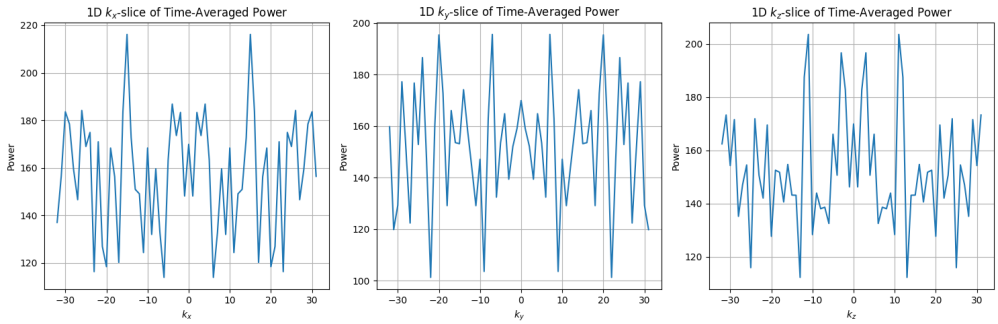

FIGURE 3. The resulting plots for each dimension show a few possible areas where the submarine's frequencies may be found but nothing conclusive. Further analysis is necessary to find the frequencies.

(4) **Identify Peak Frequency for each Dimension**
When we divide Figure 3 by the average noise across all dimensions, hence referred to as the 'Noise Floor', we find that significant peaks can be identified. Furthermore, we can see they are unique and therefore a simple filtering technique should suffice to remove the majority of the noise. We use NumPy's function `np.argmax` to find the exact Fourier space coordinates for the max values which then become the offsets in our Gaussian Filter.

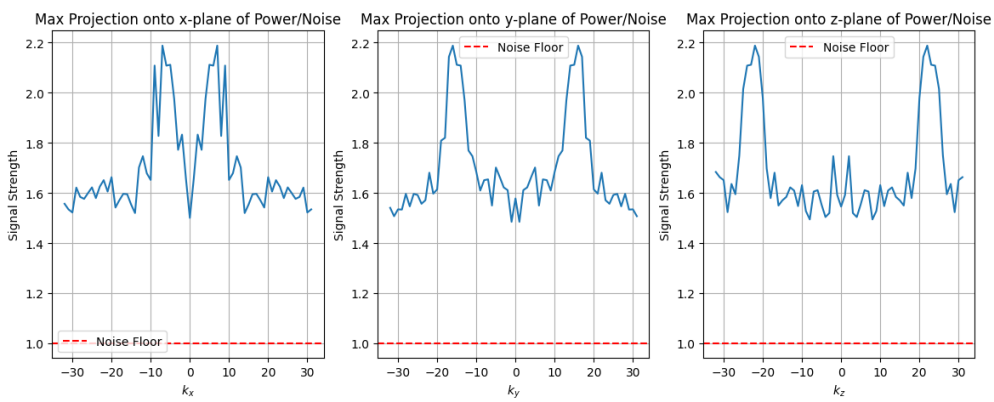

FIGURE 4. After dividing by the 'Noise Floor', clear spikes in each dimension can be identified. We can use a numpy function to find the exact value.

(5) **Create Gaussian Filter and Apply to Noisy Data**
A simple Gaussian filter is employed to maintain data around the submarine's frequency and also to eliminate noise outside its frequencies. We need to shift the filter to align with the peak frequencies identified in the previous step. We tune sigma, the width of the gaussian bump, to include as much information from the submarine as possible while balancing it against the inclusion of the white noise. After testing sigma values between 1.5 and 10, we found the best results to be for $\sigma = 5$. Our final Gaussian filter is defined as:

$$G(k_x, k_y, k_z, \sigma) = e^{-\frac{(k_x+7)^2+(k_y+16)^2+(k_z-22)^2}{2(\sigma)^2}} \tag{8}$$

(6) **Plot Results and Calculate Path**
After applying the Gaussian filter, we are able to plot the cleaned up Z-max projection of the submarine. We can validate this against the noisy data to see our cleaned up submarine follows what we were able to see in Figure 2 but with a much higher resolution. From there,

we can once again utilize `np.argmax` to find the max values in each dimension for each time step. This result is shown in Table 1

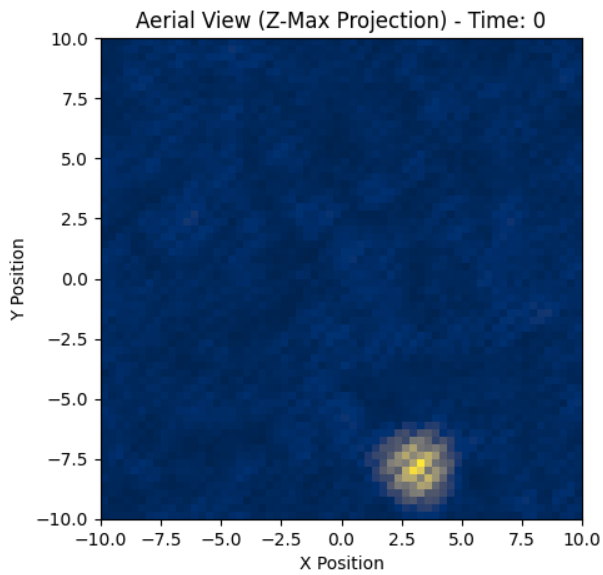

FIGURE 5. Once the data is clean, we use the same visualization function/technique as to see the difference after our first filter pass. This is the same slice we use for Figure 2.

4. Computational Results

Using the process outlined in Section 3, we identified the peak frequencies to be at $k_x = -7.0, k_y = -16.0, k_z = 22.0$. Under our assumptions, namely there were no other consistent frequency sources in our sample, we guess this is the submarine and we attempt to isolate those frequencies. These peak values were used as our offsets for the Gaussian filter as described in Equation 8. With the clean data back in the original format, we plotted its movement across time in the xy-plane and in the x,y,z space as seen in Figures 6 and 7, respectively.

TABLE 1. 3D Location of the Submarine

Time Step	**X**	**Y**	**Z**
1	3.12	-7.81	0.00
7	3.12	-6.25	2.19
14	1.88	-4.06	4.69
21	-0.62	-1.88	5.31
28	-3.44	0.00	5.62
35	-5.94	2.19	5.31
42	-6.88	4.06	3.44
49	-5.00	6.25	0.94

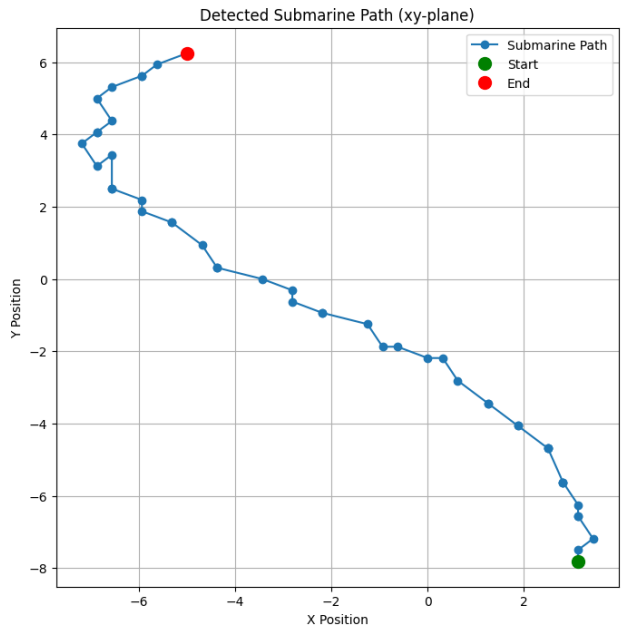

FIGURE 6. We plot the cleaned data in the xy-plane to see its movement over time, similar to seeing its movement from the top down.

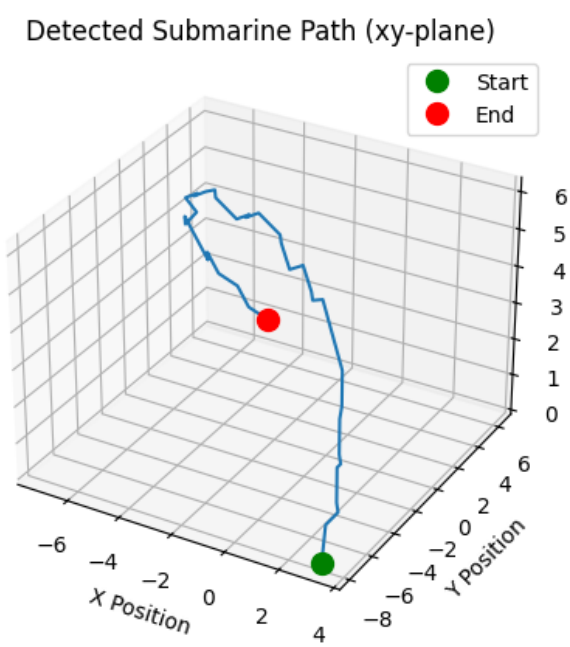

FIGURE 7. With the added z dimension, we are able to see the submarine ascend and dive while continuing to navigate in the xy plane.

5. SUMMARY AND CONCLUSIONS

In conclusion, we were able to accomplish our goals for this project, namely:

(1) Determine the frequency signature (center frequency) generated by the submarine.

(2) Design and implement a filter to extract this frequency signature in order to denoise the data and determine the path of the submarine and plot the 3D path of the submarine.
(3) Determine and plot the x, y coordinates of the submarine during the 24 hour period. This information can be used to deploy a sub-tracking aircraft to keep an eye on the submarine in the future.

Acknowledgements

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References

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